



PLC Based Height Sorting System

Factory I/O + Siemens TIA Portal / S7-PLCSIM

Project Name	Sorting by Height - Basic
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Software Used	Factory I/O, Siemens TIA Portal, S7-PLCSIM

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1. Introduction

This project presents an automated height sorting system controlled by a PLC. The system was designed and simulated using Factory I/O, while the control logic was written in Siemens TIA Portal and tested through S7-PLCSIM. The main task of the system is to move products on conveyors, detect their height using sensors, and transfer each product to the correct path automatically.

The station includes an entry conveyor, left and right transfer sections, height sensors, start/stop/reset controls, an emergency stop, indicator lamps, an alarm siren, and a counter display. The PLC reads all sensor signals and controls the actuators according to the ladder logic networks.

Project idea in one sentence

- The PLC sorts products depending on their height and counts the sorted items while providing safety and reset functions.

2. Objective

- Control a height sorting line using a PLC ladder program.
- Start and stop the system safely using physical push buttons.
- Detect product height using the high and low sensors.
- Send the product to the correct left or right path depending on its height.
- Control the conveyor entry, loading, transfer left, and transfer right actuators.
- Display start, stop, reset, warning, and alarm conditions using lamps and siren.
- Count the number of processed products using the counter output.
- Reset the system and clear the counter when required.

3. Flowchart

The following flowchart shows the basic operation sequence of the height sorting system from start-up until product counting and cycle repetition.

The PLC program is divided into three main networks. Network 1 is responsible for starting the system, operating the conveyors, detecting object height using high and low sensors, resetting the transfer outputs when the object reaches the entry sensor, and increasing the counter. Network 2 handles stop, reset, and clearing system values. Network 3 is dedicated to emergency operation by activating the warning light and alarm to place the system in a safe state.

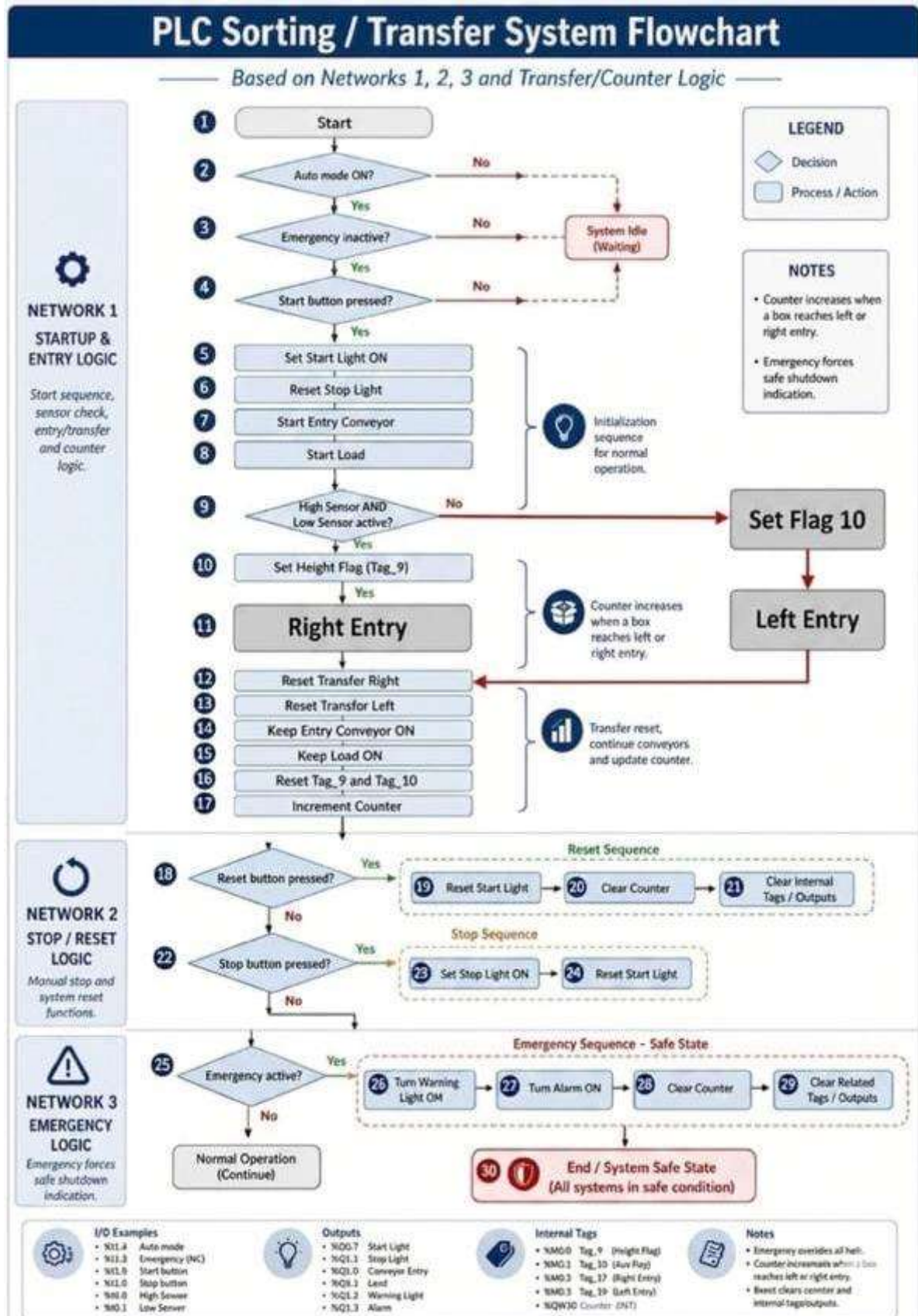


Figure 1: Flowchart of the height sorting PLC system.

4. Factory I/O Screenshots and I/O Table

The Factory I/O scene represents the physical production line. The sensors and actuators are connected to the PLC using the Siemens S7-PLCSIM driver.



Figure 2: Factory I/O driver configuration and I/O mapping.



Figure 3: General view of the height sorting conveyor system in Factory I/O.



Figure 4: Control panel showing start, stop, reset, and emergency devices.

4.1 Input Address Table

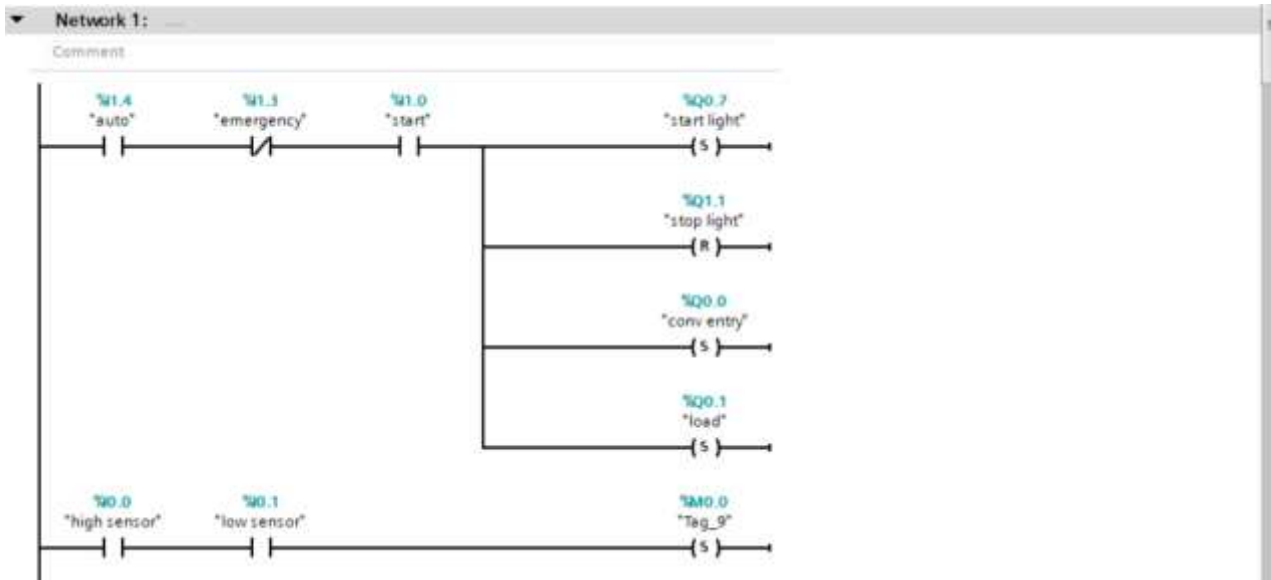
Address	Input Tag	Function
%I0.0	High sensor	Detects tall objects or the upper level of the object.
%I0.1	Low sensor	Detects the lower level/presence of the object.
%I0.2	Pallet sensor	Detects pallet/product presence in the loading area.
%I0.3	Loaded	Confirms that a product has been loaded.
%I0.4	At left entry	Detects product arrival at the left transfer entry.
%I0.5	At left exit	Detects product leaving the left path.
%I0.6	At right entry	Detects product arrival at the right transfer entry.
%I0.7	At right exit	Detects product leaving the right path.
%I1.0	Start	Starts the automatic sequence.
%I1.1	Reset	Resets system states and counter values.
%I1.2	Stop	Stops the system and activates stop indication.
%I1.3	Emergency stop	Emergency condition input for warning and siren.
%I1.4	Auto	Enables automatic operation mode.
%I1.5	Factory I/O Running	Indicates that Factory I/O simulation is running.

4.2 Output Address Table

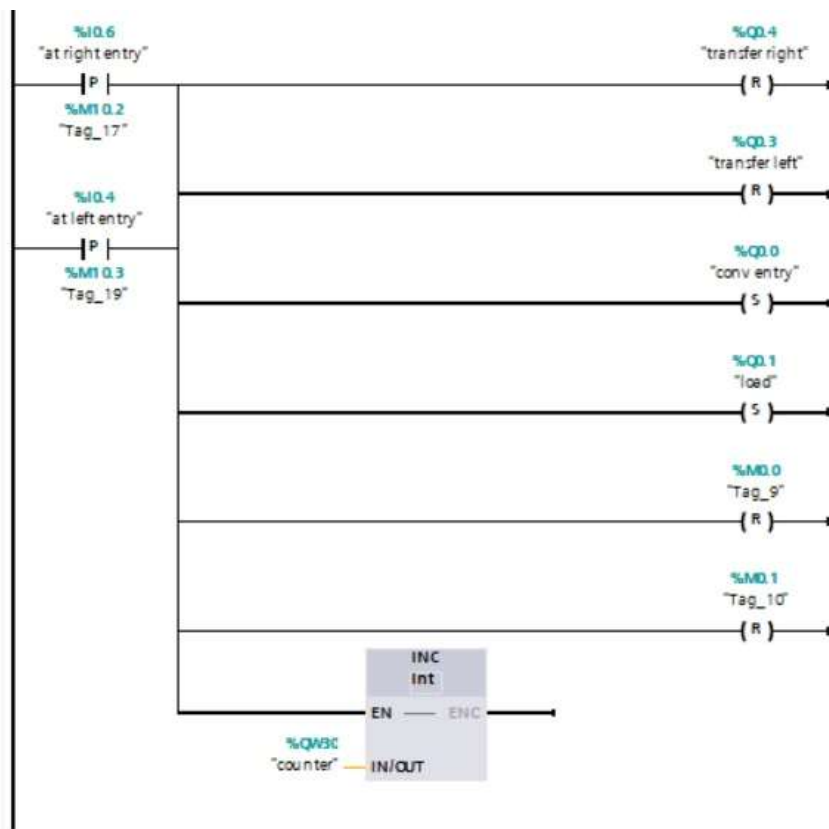
Address	Output Tag	Function
%Q0.0	Conveyor entry	Runs the entry conveyor.
%Q0.1	Load	Activates the loading mechanism.
%Q0.2	Unload	Activates the unloading mechanism.
%Q0.3	Transfer left	Moves the product toward the left path.
%Q0.4	Transfer right	Moves the product toward the right path.
%Q0.5	Conveyor left	Runs the left conveyor.
%Q0.6	Conveyor right	Runs the right conveyor.
%Q0.7	Start light	Indicates that the system is running.
%Q1.0	Reset light	Indicates reset operation.
%Q1.1	Stop light	Indicates that the system is stopped.
%Q1.2	Warning light 1	Emergency warning indication.
%Q1.3	Alarm Siren 1	Emergency audible alarm.
%QW30	Counter	Displays the number of processed products.

5. Ladder Diagram Screenshots

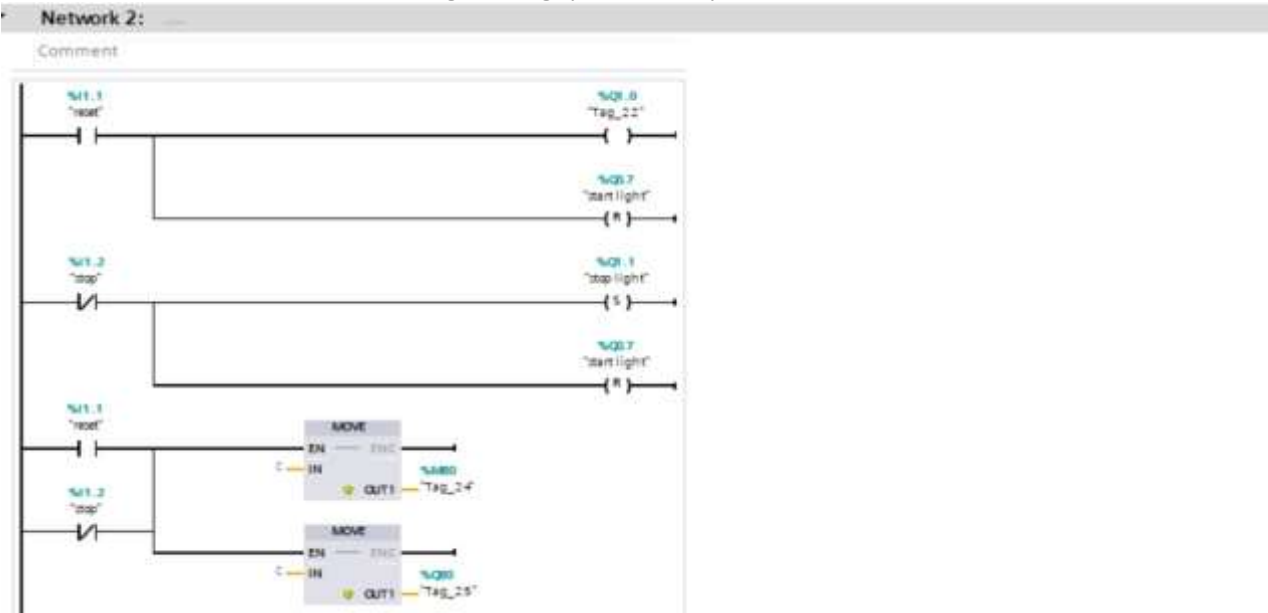
This section includes screenshots of the ladder diagram networks used in the TIA Portal PLC program.



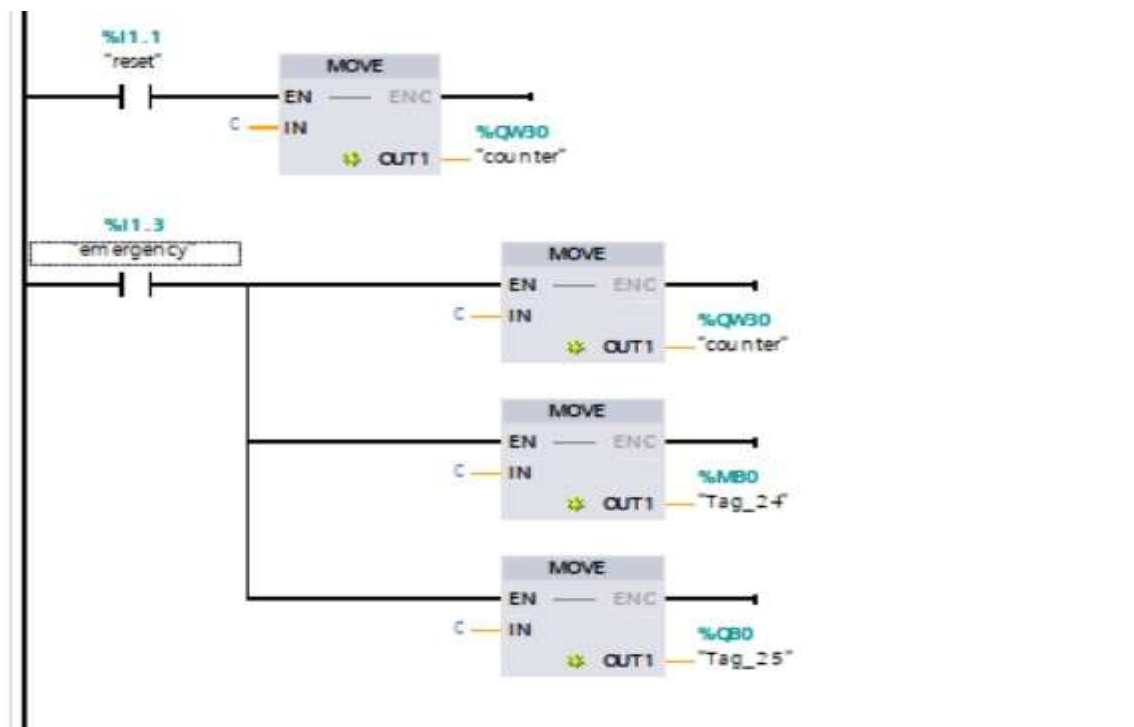
Network 1 - automatic start, start light, entry conveyor, loading, and height memory.



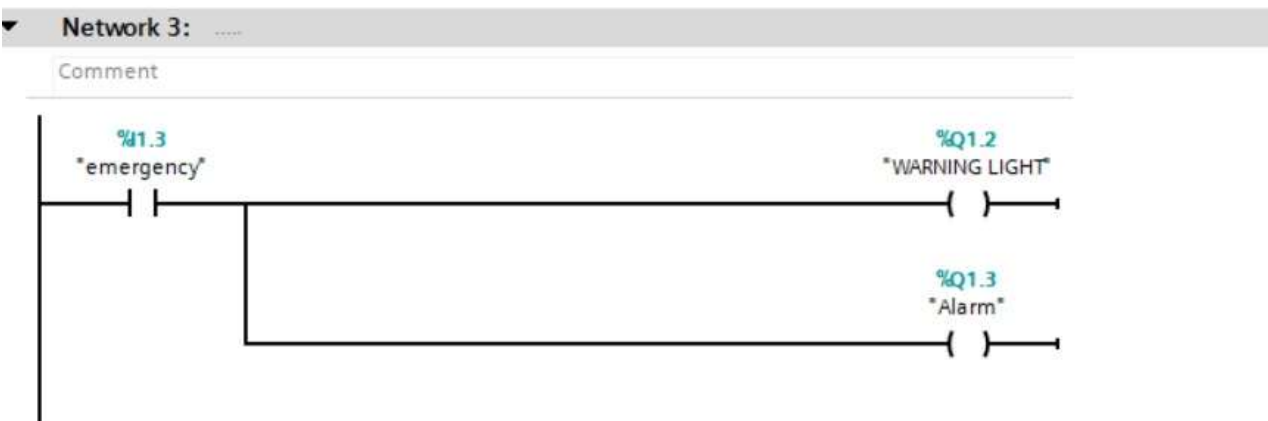
Sorting completion logic, transfer reset, loading restart, and counter increment.



Network 2 - reset/stop control and clearing of internal/output bytes.



Reset and emergency clearing logic for counter and status/output bytes



Network 3 - emergency warning light and alarm siren.

6. Ladder Diagram Code / PLC Program Networks

This section explains the ladder logic network by network. This is the part that should be written in the report in addition to the screenshots, because it shows how the PLC program works.

Network 1: Automatic Start and Height Detection Memory

This network starts the system when Auto mode is active, the emergency condition is clear, and the Start push button is pressed.

Main Condition	PLC Action
%I1.4 Auto AND NOT %I1.3 Emergency AND %I1.0 Start	SET %Q0.7 Start light, RESET %Q1.1 Stop light, SET %Q0.0 Conveyor entry, SET %Q0.1 Load
%I0.0 High sensor AND %I0.1 Low sensor	SET %M0.0 Tag_9 as height detection memory

Explanation: When the operator presses Start in Auto mode, the PLC turns on the start indicator and enables the entry conveyor and loading output. The stop light is reset. When both height sensors detect the object, the internal memory bit is set to indicate that the product is a high/tall item.

Transfer Completion and Counter Increment

This network detects when the product reaches the left or right entry position and then prepares the system for the next product.

Main Condition	PLC Action
Positive edge at %I0.6 At right entry	RESET %Q0.4 Transfer right
Positive edge at %I0.4 At left entry	RESET %Q0.3 Transfer left
Product reached a sorting entry	SET %Q0.0 Conveyor entry and SET %Q0.1 Load
Product reached a sorting entry	RESET internal memory bits %M0.0 and %M0.1
Product reached a sorting entry	Increment %QW30 Counter

Explanation: When the product reaches the selected path, the PLC stops the active transfer command. Then it restarts the entry conveyor and loading output to allow the next product to enter the system. The memory bits are cleared and the counter is increased by one.

Network 2: Reset and Stop Control

This network handles the Reset and Stop push buttons and clears internal/output data using MOVE instructions.

Main Condition	PLC Action
%I1.1 Reset	Turn on %Q1.0 Reset light and reset %Q0.7 Start light
%I1.2 Stop	SET %Q1.1 Stop light and RESET %Q0.7 Start light
Reset/Stop condition	MOVE constant value to %MB0 and %QB0 to clear memory/output byte values

Explanation: The Reset button is used to return the system to a safe initial state. The Stop button turns on the stop indication and turns off the start light. The MOVE blocks are used to clear selected memory and output bytes so that previous commands do not remain active.

Counter and Output Clearing

This network resets the counter and clears memory/output bytes during reset or emergency conditions.

Main Condition	PLC Action
%I1.1 Reset	MOVE zero/constant value to %QW30 Counter
%I1.3 Emergency stop active	MOVE zero/constant value to %QW30 Counter
%I1.3 Emergency stop active	MOVE zero/constant value to %MB0 and %QB0

Explanation: This logic is used to clear the counter and remove stored commands when Reset or Emergency is used. It prevents the machine from restarting with old memory states.

Network 3: Emergency Alarm

This network activates the safety indication when the emergency stop input is active.

Main Condition	PLC Action
%I1.3 Emergency stop active	Turn on %Q1.2 Warning Light 1
%I1.3 Emergency stop active	Turn on %Q1.3 Alarm Siren 1

Explanation: If the emergency input becomes active, the PLC immediately switches on the warning lamp and the alarm siren. This makes the fault condition visible and audible for the operator.

7. Equivalent SCL Code Representation

Note: The actual PLC program was implemented using Ladder Logic in TIA Portal. The SCL code below is only an equivalent text-based representation for documentation and discussion; it describes the same control idea in code form.

Main symbolic names used in the SCL representation:

Start = %I1.0, Reset = %I1.1, Stop = %I1.2, Emergency_Stop = %I1.3, Auto = %I1.4, High_Sensor = %I0.0, Low_Sensor = %I0.1, At_Left_Entry = %I0.4, At_Right_Entry = %I0.6, Conveyor_Entry = %Q0.0, Load = %Q0.1, Transfer_Left = %Q0.3, Transfer_Right = %Q0.4, Start_Light = %Q0.7, Reset_Light = %Q1.0, Stop_Light = %Q1.1, Warning_Light = %Q1.2, Alarm_Siren = %Q1.3, Counter = %QW30.

```
// Equivalent SCL representation of the Ladder Logic
// Actual project implementation: Ladder Logic networks in TIA Portal

// Network 1: Automatic start and loading
IF Auto AND NOT Emergency_Stop AND Start THEN
    Start_Light := TRUE;
    Stop_Light := FALSE;
    Conveyor_Entry := TRUE;
    Load := TRUE;
END IF;

// Height detection memory
IF High_Sensor AND Low_Sensor THEN
    Height_Memory := TRUE;
END_IF;

// Network 2: Stop and reset control
IF Stop THEN
    Stop_Light := TRUE;
    Start_Light := FALSE;
    Conveyor_Entry := FALSE;
    Load := FALSE;
END_IF;

IF Reset THEN
    Reset_Light := TRUE;
    Start_Light := FALSE;
    Counter := 0;
    Height_Memory := FALSE;
    Transfer_Left := FALSE;
    Transfer_Right := FALSE;
END IF;

// Network 3: Emergency alarm and safe state
IF Emergency_Stop THEN
    Warning_Light := TRUE;
    Alarm_Siren := TRUE;
    Conveyor_Entry := FALSE;
    Load := FALSE;
    Transfer_Left := FALSE;
    Transfer_Right := FALSE;
    Counter := 0;
ELSE
    Warning_Light := FALSE;
    Alarm_Siren := FALSE;
END_IF;

// Transfer completion and counter increment
IF At_Right_Entry Rising THEN
    Transfer_Right := FALSE;
    Conveyor_Entry := TRUE;
    Load := TRUE;
    Height_Memory := FALSE;
    Counter := Counter + 1;
END IF;

IF At_Left_Entry Rising THEN
    Transfer_Left := FALSE;
    Conveyor_Entry := TRUE;
    Load := TRUE;
    Height_Memory := FALSE;
    Counter := Counter + 1;
END_IF;
```

SCL Code Explanation

This SCL section documents the same idea of the Ladder program. Auto mode, Start, Stop, Reset, and Emergency inputs control the main sequence. The height sensors create the detection memory, the transfer-entry sensors finish the sorting cycle, and the counter is incremented after each completed product. During emergency, the PLC activates the warning light and siren and clears the active outputs to move the system to a safe state.

8. Operating Sequence

1. The operator opens Factory I/O, selects the Siemens S7-PLCSIM driver, and connects the simulation to the PLC program.
2. Auto mode is enabled and the emergency stop must be released.
3. The operator presses the Start button on the control panel.
4. The PLC turns on the start light, starts the entry conveyor, and activates the loading output.
5. The product moves toward the high and low sensors.
6. The PLC checks the sensor condition to determine the product height.
7. According to the height result, the product is transferred to the correct path.
8. When the product reaches the left or right entry sensor, the transfer output is reset.
9. The counter value is incremented and the system becomes ready for the next product.
10. If Stop is pressed, the system stops and the stop light turns on.
11. If Reset is pressed, the counter and selected memory/output states are cleared.
12. If Emergency is activated, the warning light and alarm siren turn on and the system is moved to a safe state.

9. Conclusion

The project successfully demonstrates a PLC controlled height sorting system. The ladder program controls the conveyors and transfer mechanisms, reads the height and position sensors, operates the control panel indicators, increments the product counter, and handles reset and emergency conditions. This type of logic is commonly used in industrial automation for sorting, packaging, and material handling applications.